

Define Problem Statements

Date	30 June 2026
Team ID	
Project Name	Rising Waters - Flood Hazard Forecasting System
Maximum Marks	2 Marks

1. The Localized Flood Susceptibility Challenge

During monsoon cycles, storm systems trigger rapid localized flash floods. Regional meteorology centers record values in Excel sheets, but hydrodynamic simulations take hours to calculate watershed overflow risks. Ground-level relief coordinators lack access to instantaneous, mobile-friendly classification resources to issue prompt warnings.

2. Hydrological Threshold Baselines

The system isolates risk boundaries based on historical trends:

- Safe Status: Combined winter/spring rainfall matches baseline loads; Jun-Sep Rainfall remains under 2425.64 mm.
- Hazard Warning Status: Cloud Cover exceeds 60% and monsoon rainfall crosses 2425.64 mm, indicating heavy convective load and high watershed flood risks.